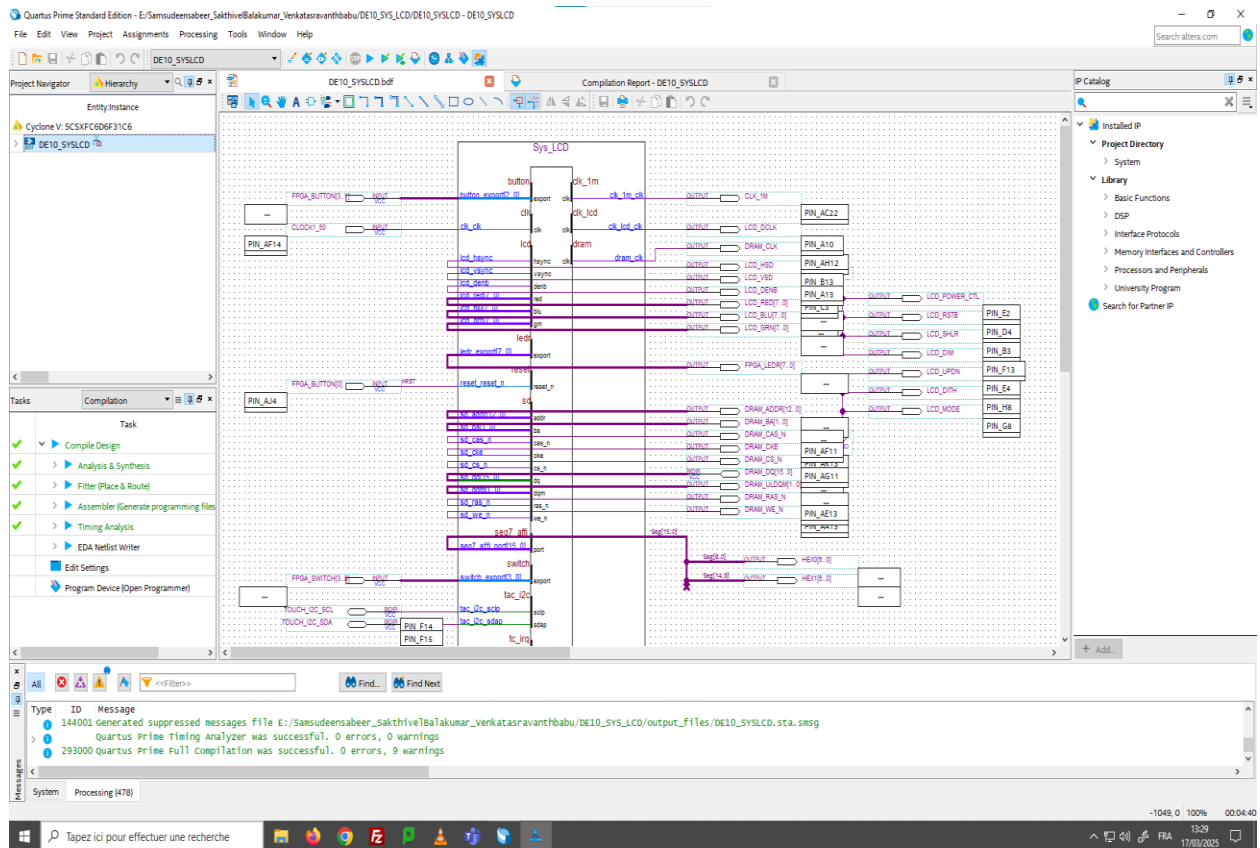


SAKTHIVEL BALAKUMAR - 63957

SAMSUDEEN SABEER RAHMAN - 63920

MEKA VENKATA SRAVANTH BABU – 63919

LAB4 – GRAPHICAL SCREEN HANDS ON THE SYSTEM



- First step is that a new project is created with “hello world” template is tested.
- Then this hello world.c file code replaced with the contents of “test.c” file.

Code of test.c file:

```
#include <stdio.h>

#include "DE10Afi_MeLib.h"
```

```

void App_UserLoop(void);

#define COLOR_BGR_FOND 0x00FF3F3F

int main()

//-----

//-----

{

    int nerr;        uint16 Version;

    printf("\n ----- Hello from Nios II/e. Test LCD COLOR ----- \n");

    DE10Lcd_Initialize(COLOR_BGR_FOND, &Version);

    printf("\n    --- LCD Hard IP version read = 0x%04X \n", Version & 0x0FFFF);

    nerr = DE10Sys_InitInterrupt();           // init des interruptions boutons, switches
    nerr += DE10Sys_InitTactile( );           // init des interruptions ecran tactile
                                           //

    if (nerr == 0) {
        DE10Sys_AffiEnable();
        //Affi_Pattern();
        App_TempoStart(2000);           // Programmer une tempo de 2 secondes pour ...
                                           // ... utiliser la fonction App_TempoWait()

        App_UserLoop();
    }
    else

        printf("\n ++++++ erreur d'initialisation des interruptions ++++++ \n\n");

    printf("\n ----- Fin de l'application ----- \n");

    return 0;

```

```
}
```

```
//----- Fonctions de la librairie -----
```

```
//          color = 0x00BBGGRR      (BB=Blue, GG=Green, RR=Red)
```

```
//          xd = pixel start (horizontal axis), yd = line start (vertical axis)
```

```
// int  DE10Affi_ReadPixel(  int xd, int yd,          int*color);
```

```
// int  DE10Affi_WritPixel(  int xd, int yd,          int color);
```

```
// int  DE10Affi_ClearZone(  int xd, int yd, int nbpix, int nblin, int color);
```

```
// int  DE10Affi_DrawVerLine( int xd, int yd, int nblin,      int color);
```

```
// int  DE10Affi_DrawHorLine( int xd, int yd, int nbpix,      int color);
```

```
// int  DE10Affi_DrawObliqLine(int xd, int yd, int nbpix,      int color, int pente);
```

```
//-----
```

```
void  App_UserLoop(void)
```

```
//-----
```

```
//-----
```

```
{
```

```
    short i, j, pixdeb, lindeb, nbpix, nblin;
```

```
    int color1; //color2, color3, color4;
```

```
    pixdeb = 100; nbpix = 200;
```

```
    lindeb = 50;  nblin = 100;
```

```
    color1 = 0x000000ff;
```

```
    while (1) {
```

```
        for (j = 0; j < nblin; j++) {
```

```
            for (i = 0; i < nbpix; i++) {
```

```
                DE10Affi_WritPixel(pixdeb+i, lindeb+j, color1);
```

```
            }
```

```
        }
```

```
        App_TempoWait();
```

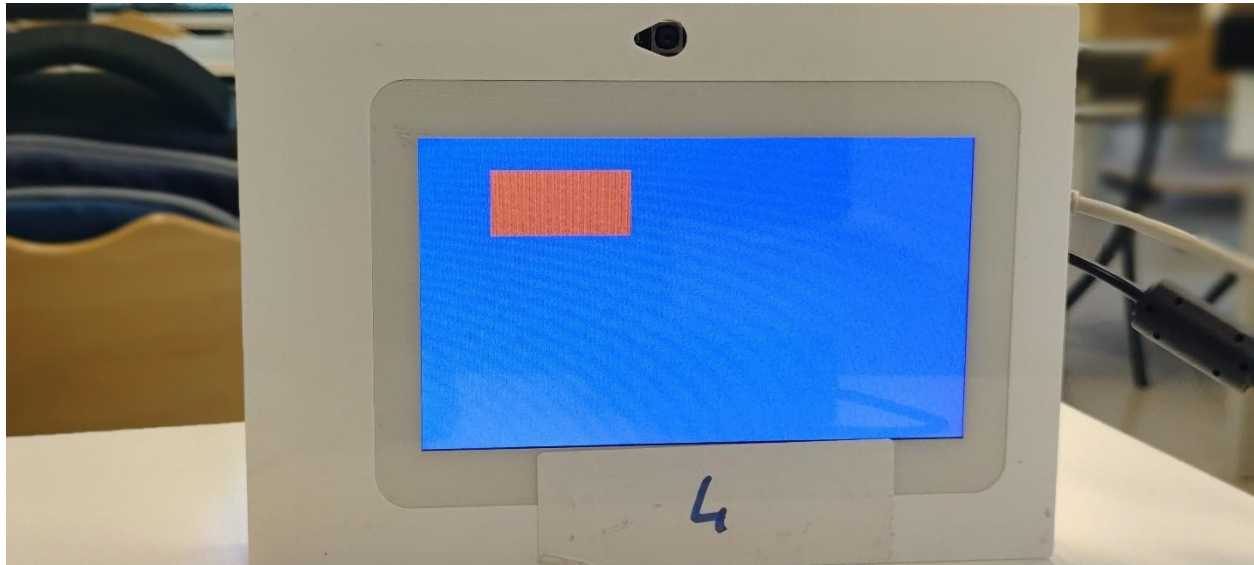
```

    }
}

```

OUTPUT:

This code is executed and a small red square is displayed on the screen.



Program to display French flag:

```

#include    <stdio.h>
#include    "DE10Afi_MeLib.h"
void  App_UserLoop(void);
#define    COLOR_BGR_FOND    0x00FF3F3F
int      main()
//-----
//-----
{
    int nerr;    uint16 Version;

    printf("\n ----- Hello from Nios II/e. Test LCD COLOR ----- \n");

    DE10Lcd_Initialize(COLOR_BGR_FOND, &Version);

```

```

printf("\n      --- LCD Hard IP version read = 0x%04X \n", Version & 0x0FFFF);

nerr = DE10Sys_InitInterrupt();          // init des interruptions boutons, switchs
nerr += DE10Sys_InitTactile( );          // init des interruptions ecran tactile
                                         //

if (nerr == 0) {
    DE10Sys_AffiEnable();
    //Affi_Pattern();
    App_TempoStart(2000);    // Programmer une tempo de 2 secondes pour ...
                             // ... utiliser la fonction
App_TempoWait()
    App_UserLoop();
}
else
    printf("\n ++++++ erreur d'initialisation des interruptions ++++++ \n\n");

printf("\n ----- Fin de l'application ----- \n");

return 0;
}

//----- Fonctions de la librairie -----
//
//          color = 0x00BBGGRR          (BB=Blue, GG=Green, RR=Red)
//
//          xd = pixel start (horizontal axis), yd = line start (vertical axis)
// int  DE10Affi_ReadPixel(  int xd, int yd,          int*color);
// int  DE10Affi_WritPixel(  int xd, int yd,          int color);
// int  DE10Affi_ClearZone(  int xd, int yd, int nbpix, int nblin, int color);
// int  DE10Affi_DrawVerLine( int xd, int yd, int nblin,      int color);
// int  DE10Affi_DrawHorLine( int xd, int yd, int nbpix,      int color);

```

```
// int  DE10Affi_DrawObliqLine(int xd, int yd, int nbpix,      int color, int pente);  
//-----
```

```
void  App_UserLoop(void)
```

```
//-----  
//-----
```

```
{  
    short i, j, pixdeb, lindeb, nbpix, nblin;  
    int blue_color = 0x00FF0000; // Blue color in 0x00BBGGRR format  
    int white_color = 0x00FFFFFF; // White color  
    int red_color = 0x000000FF; // Red color in 0x00BBGGRR format
```

```
    // Define the dimensions of the flag  
    pixdeb = 100; // Starting x-coordinate  
    nbpix = 300; // Width of the flag (3 stripes of 100 pixels each)  
    lindeb = 50; // Starting y-coordinate  
    nblin = 200; // Height of the flag
```

```
    // Calculate the width of each stripe  
    int stripe_width = nbpix / 3; // 3 equal stripes
```

```
    while (1) {  
        // Draw the blue stripe (left)  
        for (j = 0; j < nblin; j++) {  
            for (i = 0; i < stripe_width; i++) {  
                DE10Affi_WritPixel(pixdeb + i, lindeb + j, blue_color);  
            }  
        }  
    }
```

```
    // Draw the white stripe (middle)
```

```

for (j = 0; j < nblin; j++) {
    for (i = stripe_width; i < stripe_width * 2; i++) {
        DE10Affi_WritPixel(pixdeb + i, lindeb + j, white_color);
    }
}

// Draw the red stripe (right)
for (j = 0; j < nblin; j++) {
    for (i = stripe_width * 2; i < nbpix; i++) {
        DE10Affi_WritPixel(pixdeb + i, lindeb + j, red_color);
    }
}

// Infinite loop to keep the flag displayed
App_TempoWait();
}
}

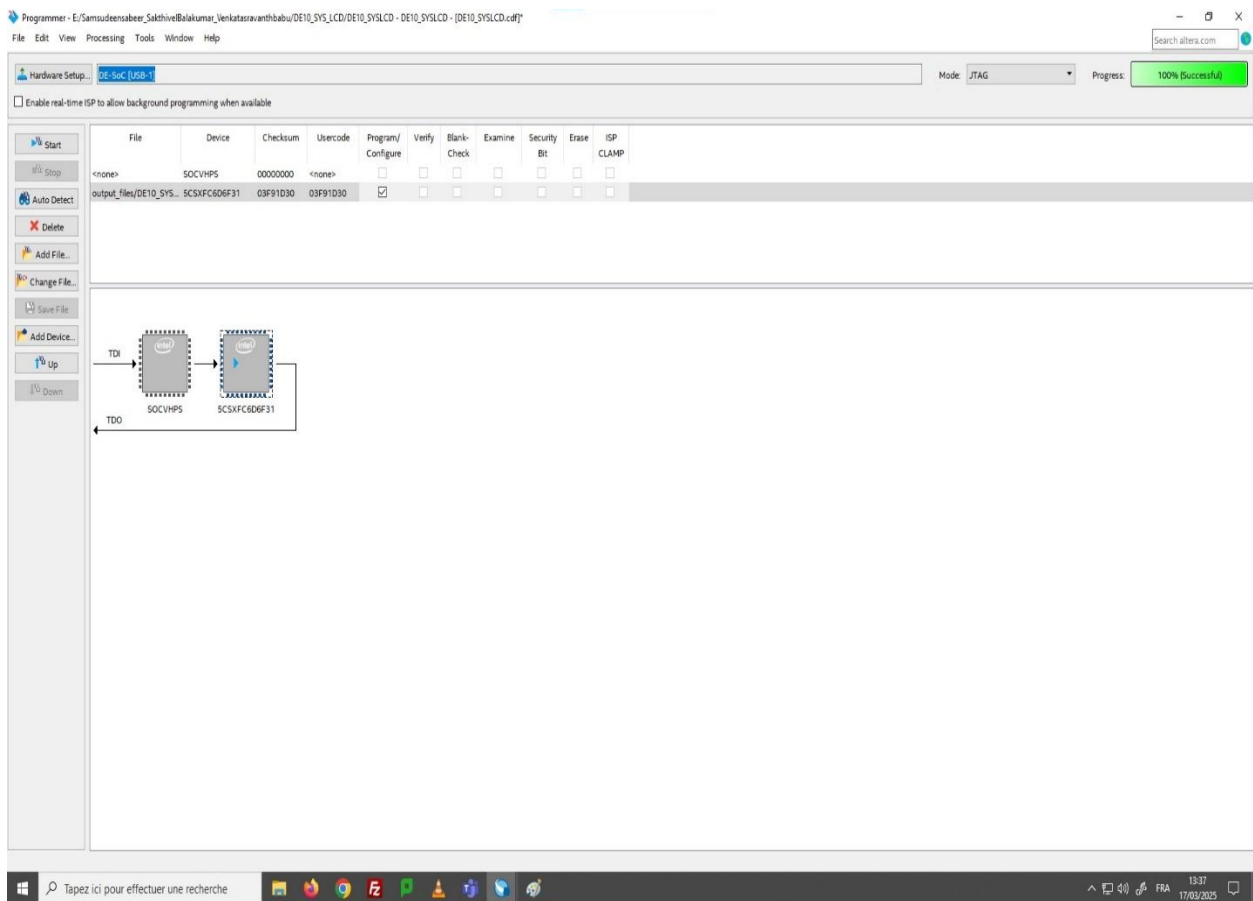
```

OUTPUT:

This code is executed to display the French flag.



Output of 100% progress successful:



Q1) Explain how the graphical screen code each pixel:

First we need to understand the pixel format, the LCD uses a 0x00BBGGRR format. 8 bits for each colour (blue, green and red).

This code is indicated in hexadecimal format.

Colour indication for each:

Blue: 0x00FF0000;

White: 0x00FFFFFF;

Red: 0x000000FF;

Pixel coordinate of the flag:

The flag displayed at pixel coordinate (100,50) on the screen. Width is 300 pixels and height is 200 pixels.

The flag contains 3 vertical stripes, each are 100 pixels wide.

Blue strip (Left side):

```
// Draw the blue stripe (left)
```

```
    for (j = 0; j < nblin; j++) {  
        for (i = 0; i < stripe_width; i++) {  
            DE10Affi_WritPixel(pixdeb + i, lindeb + j, blue_color);  
        }  
    }
```

White strip (Middle):

```
// Draw the white stripe (middle)
```

```
    for (j = 0; j < nblin; j++) {  
        for (i = stripe_width; i < stripe_width * 2; i++) {  
            DE10Affi_WritPixel(pixdeb + i, lindeb + j, white_color);  
        }  
    }
```

Red strip (Right side):

```
// Draw the red stripe (right)
```

```
    for (j = 0; j < nblin; j++) {  
        for (i = stripe_width * 2; i < nbpix; i++) {  
            DE10Affi_WritPixel(pixdeb + i, lindeb + j, red_color);  
        }  
    }
```

Explanation:

Each pixel on the screen is set using the function :

```
DE10Affi_WritPixel(x, y, color);
```

The outer for loop [for (j = 0; j < nblin; j++)] moves vertically row by row.

The inner for loop [for (i = start; i < end; i++)] moves horizontally pixel by pixel in that row.

Each pixel in the region is colored using `DE10Affi_WritPixel(x, y, color)` ;

This code works for the tricolor flag in this format:

Left blue strip – iterates over the first third of the width (0 to `stripe_width`) and blue color comes.

Middle white strip- iterates over the next third (`stripe_width` to `2 * stripe_width`) and colors it white.

Right Red strip – iterates over the last third (`2 * stripe_width` to `nbpix`).

General concept about graphical screen displays each pixel:

First pixel representation

Actually the screen is a 2D array of pixels arranged in (x , y) coordinates.

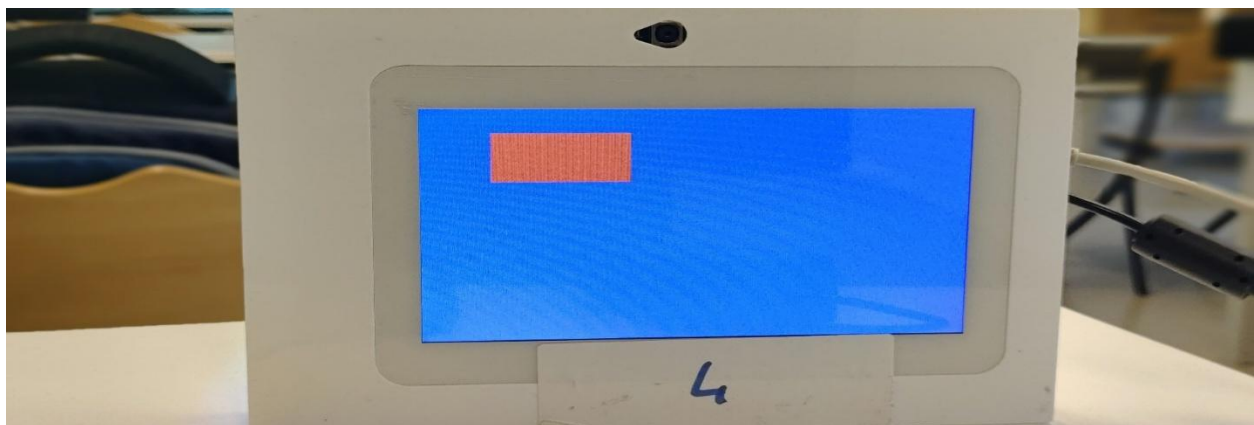
Colour encoding

Colour is coded by changing the hexadecimal code (For eg:0x000000ff)

Then drawing the flag in code

Here we are dividing the screen into sections to create flag like (vertical or horizontal stripes, symbols etc.)

For the small red square just we are taking for loop only one time and determining the values to display red square.



Q2) Program to display a yellow square with 2 different intensities:

Code:

```
#include    <stdio.h>
#include    "DE10Afi_MeLib.h"
void  App_UserLoop(void);
#define    COLOR_BGR_FOND    0x00FF3F3F
int    main()
//-----
//-----
{
    int nerr;    uint16 Version;

    printf("\n ----- Hello from Nios II/e. Test LCD COLOR ----- \n");

    DE10Lcd_Initialize(COLOR_BGR_FOND, &Version);

    printf("\n    --- LCD Hard IP version read = 0x%04X \n", Version & 0x0FFFF);

    nerr = DE10Sys_InitInterrupt();    // init des interruptions boutons, switches
    nerr += DE10Sys_InitTactile( );    // init des interruptions ecran tactile
                                     //

    if (nerr == 0) {
        DE10Sys_AffiEnable();
        //Affi_Pattern();
        App_TempoStart(2000);    // Programmer une tempo de 2 secondes pour ...
                                // ... utiliser la fonction App_TempoWait()

        App_UserLoop();
    }
    else
```

```

        printf("\n ++++++ erreur d'initialisation des interruptions ++++++ \n\n");

printf("\n ----- Fin de l'application ----- \n");

return 0;
}

//----- Fonctions de la librairie -----
//
//          color = 0x00BBGGRR      (BB=Blue, GG=Green, RR=Red)
//
//          xd = pixel start (horizontal axis), yd = line start (vertical axis)
// int  DE10Affi_ReadPixel(  int xd, int yd,          int*color);
// int  DE10Affi_WritPixel(  int xd, int yd,          int color);
// int  DE10Affi_ClearZone(  int xd, int yd, int nbpix, int nblin, int color);
// int  DE10Affi_DrawVerLine( int xd, int yd, int nblin,      int color);
// int  DE10Affi_DrawHorLine( int xd, int yd, int nbpix,      int color);
// int  DE10Affi_DrawObliqLine(int xd, int yd, int nbpix,      int color, int pente);
//-----
void  App_UserLoop(void)
//-----
//-----
{
    short i, j, pixdeb, lindeb, nbpix, nblin;
    int yellow_light = 0x0080FF80; // Light yellow (more green, less red)
    int yellow_dark = 0x0000A5A5;  // Dark yellow (more red, less green)

    // Define the dimensions of the square
    pixdeb = 100; // Starting x-coordinate
    nbpix = 200;  // Width of the square
    lindeb = 50;  // Starting y-coordinate
    nblin = 200;  // Height of the square

```

```

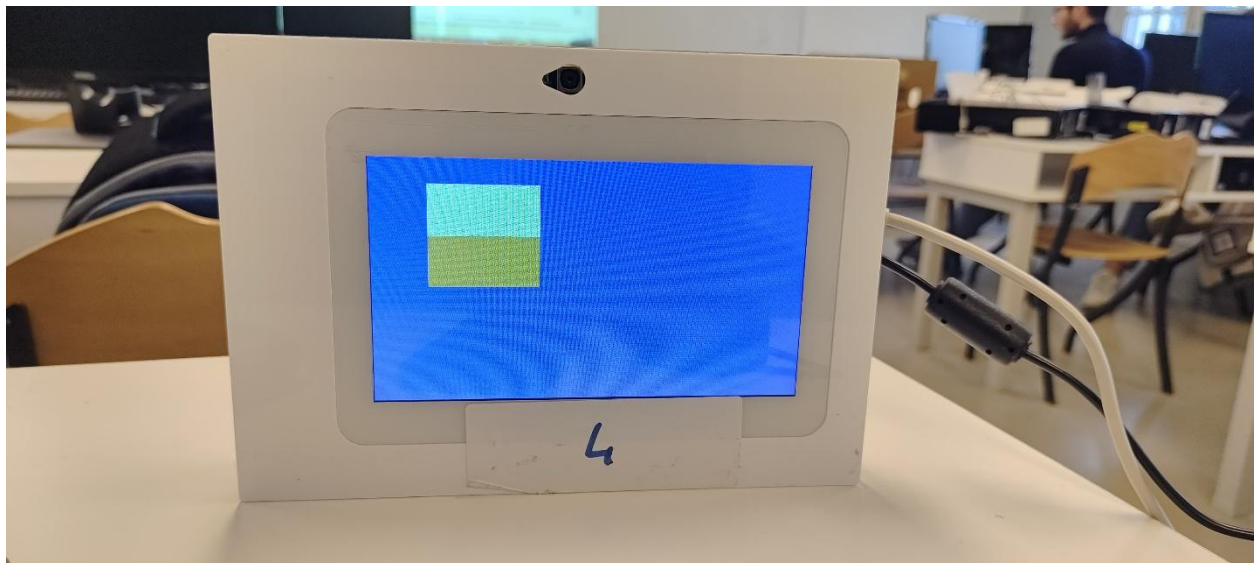
while (1) {
    // Draw the top half of the square with light yellow
    for (j = 0; j < nblin / 2; j++) {
        for (i = 0; i < nbpix; i++) {
            DE10Affi_WritPixel(pixdeb + i, lindeb + j, yellow_light);
        }
    }

    // Draw the bottom half of the square with dark yellow
    for (j = nblin / 2; j < nblin; j++) {
        for (i = 0; i < nbpix; i++) {
            DE10Affi_WritPixel(pixdeb + i, lindeb + j, yellow_dark);
        }
    }

    // Infinite loop to keep the square displayed
    App_TempoWait();
}

```

Output:



Q3) The lena256.h to display an image of Lena:

Code:

```
/*
 * "Hello World" example with Lena image in original colors with brightness and contrast adjustment.
 */

#include <stdio.h>
#include "DE10Afi_MeLib.h"

#include "Lena256.h"
#define SIZE_LENA 256

void App_UserLoop(void);
void Affi_Lena(short pixdeb, short lindeb, int brightness, float contrast);

#define COLOR_BGR_FOND 0x00FF3F3F

int main()
{
    int nerr; uint16 Version;

    printf("\n ----- Hello from Nios II/e. Test LCD COLOR ----- \n");

    DE10Lcd_Initialize(COLOR_BGR_FOND, &Version);

    printf("\n --- LCD Hard IP version read = 0x%04X \n", Version & 0xFFFF);

    nerr = DE10Sys_InitInterrupt();
    nerr += DE10Sys_InitTactile();

    if (nerr == 0) {
```

```

        DE10Sys_AffiEnable();
        App_TempoStart(2000);
        Affi_Lena(50, 50, 30, 1.2); // Adjust brightness and contrast here
    }
    else
        printf("\n ++++++ erreur d'initialisation des interruptions ++++++ \n\n");

    printf("\n ----- Fin de l'application ----- \n");

    return 0;
}

int AdjustValue(int value, int brightness, float contrast) {
    int result = (int)((value - 128) * contrast + 128) + brightness;
    if (result > 255) result = 255;
    if (result < 0) result = 0;
    return result;
}

void Affi_Lena(short pixdeb, short lindeb, int brightness, float contrast)
{
    int i, j, idc, color;        int pixel;
    int red, green, blue;

    while (1) {
        idc = 0;
        for (j = 0; j < SIZE_LENA; j++) {
            for (i = 0; i < SIZE_LENA; i++) {
color = Lena256[idc];

// Extract RGB components

```

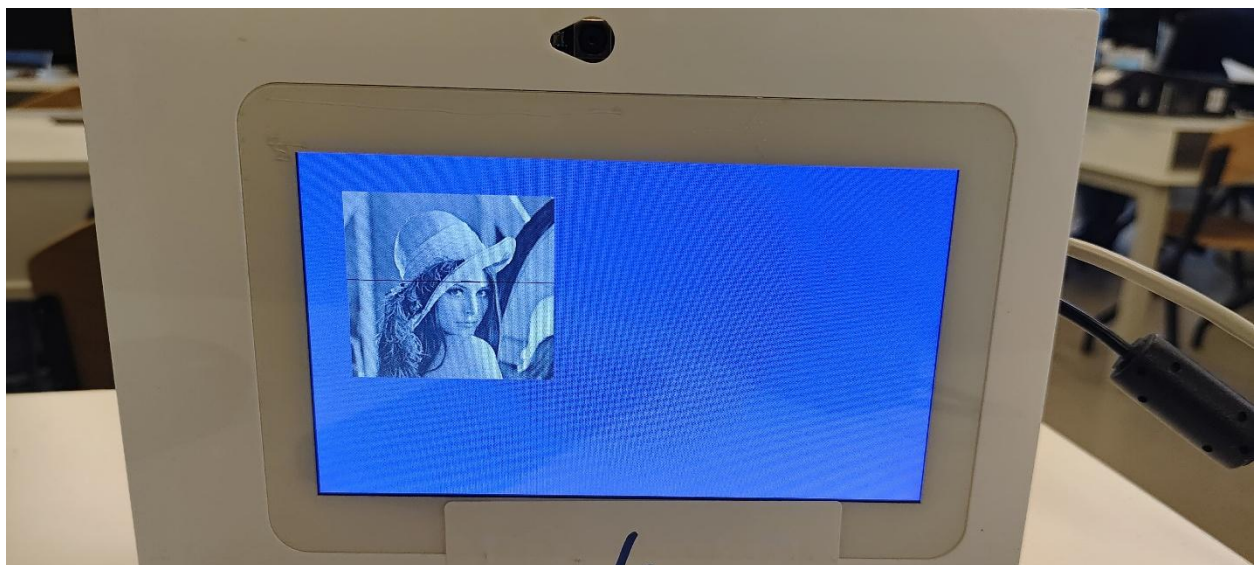
```

        blue = color & 0xFF;
        green = color & 0xFF;
        red = color & 0xFF;
        /*
// Apply brightness and contrast adjustments
        red = AdjustValue(red, brightness, contrast);
        green = AdjustValue(green, brightness, contrast);
        blue = AdjustValue(blue, brightness, contrast);
*/

        // Combine adjusted components
        color = (blue << 16) | (green << 8) | red;
        DE10Affi_WritPixel(pixdeb + i, lindeb + j, color);
        idc++;
    }
}
App_TempoWait();
}
}

```

OUTPUT:



Q4) If we have a RGB image, how can we obtain a gray level image ?

In a RGB representation, any image is composed of pixels made of three components Red(R),Green(G),Blue(B)

In order to convert a RGB image to gray level image we need to do some calculations like

$$\text{Gray} = 0.299 * R + 0.587 * G + 0.114 * B$$

All the three RGB components, made as Gray for each pixel.

Steps to Convert RGB to Grayscale:

1. **Extract the R, G, and B channels** from the RGB image.
2. **Apply the weighted average** to each pixel using the formula above.
3. **Combine the results** to form the grayscale image.